

# SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

*Symbiosis International (Deemed University)*

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**Experiment No.: 8      Name: Vansh Mahant      PRN: 24070521038      Section: A      Batch: A1**

## 1. Experiment Title:

Part A: Gym Admission Registration Form — a table-structured web form using HTML, CSS and JavaScript demonstrating multiple form controls (text, radio, select, checkbox, textarea) with real-time validation and a live character counter, built around JavaScript events (input, blur, focus, change, submit, reset).

Part B: Gym Admission Form — a compact single-file web form using HTML, CSS and JavaScript demonstrating field-level validation (name, age, email, mobile number, membership plan) using input, blur, change and submit events.

## 2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, JavaScript

## 3. Experiment Program Code:

### Part A — Gym Admission Registration Form

[index.html](#)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Gym Admission Registration Form - Practical 8</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="form-container">
    <h2>Gym Admission Registration Form</h2>
    <p class="subtitle">Practical 8: Form Validation & JavaScript Events</p>

    <form id="gymRegistrationForm">
      <table class="gym-table">
        <thead>
          <tr>
            <th class="col-header">Name</th>
            <th class="col-header">Value</th>
          </tr>
        </thead>
        <tbody>
          <!-- Row 1: Name -->
          <tr>
            <td class="label-cell">
              <label for="fullName">Name</label>
            </td>
            <td class="input-cell">
              <input type="text" id="fullName" name="fullName" placeholder="Enter
your full name">
              <span id="nameError" class="error-msg"></span>
            </td>
          </tr>

          <!-- Row 2: Sex -->
          <tr>
```

```

        <td class="label-cell">
            <label>Sex</label>
        </td>
        <td class="input-cell">
            <div class="radio-group">
                <label class="option-label">
                    <input type="radio" name="sex" value="Male" checked> Male
                </label>
                <label class="option-label">
                    <input type="radio" name="sex" value="Female"> Female
                </label>
            </div>
            <span id="sexError" class="error-msg"></span>
        </td>
    </tr>

    <!-- Row 3: Eye Color -->
    <tr>
        <td class="label-cell">
            <label for="eyeColor">Eye color</label>
        </td>
        <td class="input-cell">
            <select id="eyeColor" name="eyeColor">
                <option value="green" selected>green</option>
                <option value="blue">blue</option>
                <option value="brown">brown</option>
                <option value="hazel">hazel</option>
                <option value="black">black</option>
                <option value="gray">gray</option>
            </select>
            <span id="eyeColorStatus" class="info-msg"></span>
        </td>
    </tr>

    <!-- Row 4: Check all that apply -->
    <tr>
        <td class="label-cell">
            <label>Check all that apply</label>
        </td>
        <td class="input-cell">
            <div class="checkbox-group">
                <label class="option-label">
                    <input type="checkbox" name="attributes" value="Over 6 feet
tall"> Over 6 feet tall
                </label>
                <label class="option-label">
                    <input type="checkbox" name="attributes" value="Over 200
pounds"> Over 200 pounds
                </label>
            </div>
            <span id="attrCount" class="info-msg"></span>
        </td>
    </tr>

    <!-- Row 5: Describe athletic ability -->
    <tr>
        <td colspan="2" class="textarea-cell">
            <label for="athleticAbility" class="block-label">Describe your
athletic ability:</label>
            <textarea id="athleticAbility" name="athleticAbility" rows="5"
placeholder="Mention your fitness background, sports experience, or athletic
goals..."></textarea>
            <div class="char-counter">
                <span id="charCount">0</span> characters entered
            </div>
            <span id="abilityError" class="error-msg"></span>
        </td>
    </tr>

```

```

                <!-- Row 6: Submit Button -->
                <tr>
                    <td colspan="2" class="button-cell">
                        <button type="submit" id="submitBtn">Enter my information</button>
                        <button type="reset" id="resetBtn">Reset</button>
                    </td>
                </tr>
            </tbody>
        </table>
    </form>

    <!-- Result / Confirmation Display -->
    <div id="resultCard" class="result-card" style="display: none;">
        <h3>Registration Submitted Successfully!</h3>
        <div id="summaryDetails"></div>
    </div>
</div>

<script src="script.js"></script>
</body>
</html>

```

### style.css

```

/* Simple, Clean CSS for Gym Admission Registration Form (Practical 8) */

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, Helvetica, sans-serif;
    background-color: #f0f2f5;
    color: #333333;
    padding: 30px 15px;
    line-height: 1.5;
}

/* Main Container */
.form-container {
    max-width: 580px;
    margin: 0 auto;
    background: #ffffff;
    padding: 25px;
    border: 1px solid #d0d7de;
    border-radius: 8px;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.08);
}

h2 {
    text-align: center;
    color: #2c3e50;
    margin-bottom: 5px;
    font-size: 22px;
}

.subtitle {
    text-align: center;
    color: #666666;
    font-size: 13px;
    margin-bottom: 20px;
}

/* Table Style matching the practical sheet */
.gym-table {
    width: 100%;
    border-collapse: collapse;
}

```

```

    border: 1px solid #b8c4d4;
    background-color: #ffffff;
}

.gym-table th,
.gym-table td {
    border: 1px solid #b8c4d4;
    padding: 10px 14px;
    font-size: 14px;
    vertical-align: top;
}

/* Header styling */
.col-header {
    background-color: #c4d4ec;
    color: #1a2a3a;
    font-weight: bold;
    text-align: left;
    padding: 8px 14px;
}

/* Column 1: Label cell */
.label-cell {
    width: 40%;
    font-weight: 500;
    background-color: #f8fafc;
    color: #333333;
}

/* Column 2: Input cell */
.input-cell {
    width: 60%;
}

/* Form Controls */
input[type="text"],
select,
textarea {
    width: 100%;
    padding: 7px 10px;
    border: 1px solid #a0b0c0;
    border-radius: 3px;
    font-size: 14px;
    font-family: inherit;
    outline: none;
    transition: border-color 0.2s, box-shadow 0.2s;
}

input[type="text"]:focus,
select:focus,
textarea:focus {
    border-color: #3b82f6;
    box-shadow: 0 0 4px rgba(59, 130, 246, 0.3);
}

/* Radio & Checkbox options */
.radio-group,
.checkbox-group {
    display: flex;
    flex-direction: column;
    gap: 6px;
}

.option-label {
    display: flex;
    align-items: center;
    gap: 8px;
    cursor: pointer;
    font-weight: normal;
}

```

```

}

.option-label input[type="radio"],
.option-label input[type="checkbox"] {
    cursor: pointer;
    accent-color: #2563eb;
}

/* Textarea row */
.textarea-cell {
    background-color: #fbfcfe;
}

.block-label {
    display: block;
    font-weight: 600;
    margin-bottom: 8px;
    color: #2c3e50;
}

textarea {
    resize: vertical;
    min-height: 90px;
}

.char-counter {
    font-size: 12px;
    color: #777777;
    text-align: right;
    margin-top: 4px;
}

/* Action Buttons */
.button-cell {
    text-align: center;
    background-color: #f1f5f9;
    padding: 14px;
}

button {
    font-size: 14px;
    padding: 8px 18px;
    border: 1px solid #999999;
    border-radius: 4px;
    cursor: pointer;
    font-weight: 500;
    transition: background-color 0.2s, color 0.2s;
}

#submitBtn {
    background-color: #e5edf7;
    color: #1a2a3a;
    border: 1px solid #8ca6c8;
    margin-right: 8px;
}

#submitBtn:hover {
    background-color: #c9dbf2;
}

#resetBtn {
    background-color: #f3f4f6;
    color: #555555;
    border: 1px solid #ccc;
}

#resetBtn:hover {
    background-color: #e5e7eb;
}

```

```

/* Validation Messages */
.error-msg {
    display: block;
    color: #dc2626;
    font-size: 12px;
    margin-top: 4px;
    font-weight: normal;
}

.info-msg {
    display: block;
    color: #0284c7;
    font-size: 12px;
    margin-top: 4px;
}

/* Valid input indicator */
.is-valid {
    border-color: #16a34a !important;
}

.is-invalid {
    border-color: #dc2626 !important;
    background-color: #fef2f2;
}

/* Submission Result Card */
.result-card {
    margin-top: 20px;
    padding: 18px;
    background: #ecfdf5;
    border: 1px solid #a7f3d0;
    border-radius: 6px;
    color: #065f46;
}

.result-card h3 {
    font-size: 16px;
    margin-bottom: 12px;
    color: #047857;
    border-bottom: 1px dashed #a7f3d0;
    padding-bottom: 6px;
}

.summary-item {
    margin-bottom: 6px;
    font-size: 13.5px;
    color: #1f2937;
}

.summary-item strong {
    color: #065f46;
    display: inline-block;
    width: 170px;
}

```

### script.js

```

// =====
// Practical 8: Gym Admission Registration Form
// Demonstrating JavaScript Events & Form Validation
// =====

document.addEventListener("DOMContentLoaded", function () {
    // Form and input elements
    const form = document.getElementById("gymRegistrationForm");
    const fullNameInput = document.getElementById("fullName");
    const sexRadios = document.querySelectorAll('input[name="sex"]');

```

```

const eyeColorSelect = document.getElementById("eyeColor");
const attributeCheckboxes = document.querySelectorAll('input[name="attributes"]');
const athleticAbilityTextarea = document.getElementById("athleticAbility");
const charCountSpan = document.getElementById("charCount");
const resultCard = document.getElementById("resultCard");
const summaryDetails = document.getElementById("summaryDetails");

// Error message elements
const nameError = document.getElementById("nameError");
const sexError = document.getElementById("sexError");
const abilityError = document.getElementById("abilityError");
const eyeColorStatus = document.getElementById("eyeColorStatus");
const attrCount = document.getElementById("attrCount");

// -----
// 1. Validation Helper Functions
// -----

// Validate Full Name
function validateName() {
    const value = fullNameInput.value.trim();
    const nameRegex = /^[A-Za-z\s]+$/;

    if (value === "") {
        nameError.textContent = "Name is required.";
        fullNameInput.classList.add("is-invalid");
        fullNameInput.classList.remove("is-valid");
        return false;
    } else if (value.length < 3) {
        nameError.textContent = "Name must be at least 3 characters long.";
        fullNameInput.classList.add("is-invalid");
        fullNameInput.classList.remove("is-valid");
        return false;
    } else if (!nameRegex.test(value)) {
        nameError.textContent = "Name must contain only alphabets and spaces.";
        fullNameInput.classList.add("is-invalid");
        fullNameInput.classList.remove("is-valid");
        return false;
    } else {
        nameError.textContent = "";
        fullNameInput.classList.remove("is-invalid");
        fullNameInput.classList.add("is-valid");
        return true;
    }
}

// Validate Sex selection
function validateSex() {
    let isSelected = false;
    sexRadios.forEach((radio) => {
        if (radio.checked) isSelected = true;
    });

    if (!isSelected) {
        sexError.textContent = "Please select your gender.";
        return false;
    } else {
        sexError.textContent = "";
        return true;
    }
}

// Validate Athletic Ability
function validateAthleticAbility() {
    const value = athleticAbilityTextarea.value.trim();

    if (value === "") {
        abilityError.textContent = "Please describe your athletic ability.";
        athleticAbilityTextarea.classList.add("is-invalid");
    }
}

```

```

        athleticAbilityTextarea.classList.remove("is-valid");
        return false;
    } else if (value.length < 10) {
        abilityError.textContent = "Description must be at least 10 characters long.";
        athleticAbilityTextarea.classList.add("is-invalid");
        athleticAbilityTextarea.classList.remove("is-valid");
        return false;
    } else {
        abilityError.textContent = "";
        athleticAbilityTextarea.classList.remove("is-invalid");
        athleticAbilityTextarea.classList.add("is-valid");
        return true;
    }
}

// -----
// 2. JavaScript Events Demonstration
// -----

// EVENT 1: 'input' event on Full Name (real-time validation)
fullNameInput.addEventListener("input", function () {
    validateName();
});

// EVENT 2: 'blur' event on Full Name (validation on leaving field)
fullNameInput.addEventListener("blur", function () {
    validateName();
});

// EVENT 3: 'focus' event on Full Name & Textarea (visual highlight)
fullNameInput.addEventListener("focus", function () {
    if (nameError.textContent === "") {
        fullNameInput.classList.remove("is-invalid");
    }
});

// EVENT 4: 'change' event on Sex radio buttons
sexRadios.forEach((radio) => {
    radio.addEventListener("change", function () {
        validateSex();
    });
});

// EVENT 5: 'change' event on Eye Color dropdown
eyeColorSelect.addEventListener("change", function () {
    eyeColorStatus.textContent = `Selected: ${this.value}`;
    setTimeout(() => {
        eyeColorStatus.textContent = "";
    }, 2000);
});

// EVENT 6: 'change' event on Checkboxes (Attributes)
attributeCheckboxes.forEach((checkbox) => {
    checkbox.addEventListener("change", function () {
        const selectedCount = Array.from(attributeCheckboxes).filter(cb =>
cb.checked).length;
        attrCount.textContent = `${selectedCount} option(s) selected`;
        setTimeout(() => {
            attrCount.textContent = "";
        }, 2000);
    });
});

// EVENT 7: 'input' & 'keyup' events on Athletic Ability textarea (Character counter)
athleticAbilityTextarea.addEventListener("input", function () {
    const length = this.value.length;
    charCountSpan.textContent = length;
    validateAthleticAbility();
});

```



```

// EVENT 8: 'submit' event on Form
form.addEventListener("submit", function (e) {
  // Prevent default browser form submission
  e.preventDefault();

  // Perform all validations
  const isNameValid = validateName();
  const isSexValid = validateSex();
  const isAbilityValid = validateAthleticAbility();

  if (isNameValid && isSexValid && isAbilityValid) {
    // Collect Form Values
    const nameVal = fullNameInput.value.trim();
    const sexVal = document.querySelector('input[name="sex"]:checked')?.value || "Not
specified";
    const eyeColorVal = eyeColorSelect.value;

    const selectedAttributes = Array.from(attributeCheckboxes)
      .filter(cb => cb.checked)
      .map(cb => cb.value);
    const attributesVal = selectedAttributes.length > 0 ? selectedAttributes.join(", ") :
"None";

    const abilityVal = athleticAbilityTextarea.value.trim();

    // Display Summary Details
    summaryDetails.innerHTML = `
      <div class="summary-item"><strong>Full Name:</strong> ${nameVal}</div>
      <div class="summary-item"><strong>Sex:</strong> ${sexVal}</div>
      <div class="summary-item"><strong>Eye Color:</strong> ${eyeColorVal}</div>
      <div class="summary-item"><strong>Attributes:</strong> ${attributesVal}</div>
      <div class="summary-item"><strong>Athletic Ability:</strong> ${abilityVal}</div>
    `;

    resultCard.style.display = "block";
    resultCard.scrollIntoView({ behavior: "smooth" });

    alert("Admission Information submitted successfully!");
  } else {
    resultCard.style.display = "none";
    alert("Please resolve the highlighted errors before submitting.");
  }
});

// EVENT 9: 'reset' event on Form
form.addEventListener("reset", function () {
  // Clear all validation errors and states
  nameError.textContent = "";
  sexError.textContent = "";
  abilityError.textContent = "";
  eyeColorStatus.textContent = "";
  attrCount.textContent = "";
  charCountSpan.textContent = "0";

  fullNameInput.classList.remove("is-valid", "is-invalid");
  athleticAbilityTextarea.classList.remove("is-valid", "is-invalid");

  resultCard.style.display = "none";
  summaryDetails.innerHTML = "";
});
});

```

## Part B — Gym Admission Form

### INDEX.html

```

<!DOCTYPE html>
<html>

```

```

<head>
  <title>Gym Admission Form</title>
  <style>
    body{
      font-family: Arial, sans-serif;
      background:#f2f2f2;
    }
    .container{
      width:400px;
      margin:40px auto;
      padding:20px;
      background:white;
      border-radius:10px;
      box-shadow:0 0 10px gray;
    }
    h2{
      text-align:center;
    }
    label{
      display:block;
      margin-top:10px;
    }
    input, select{
      width:100%;
      padding:8px;
      margin-top:5px;
    }
    .error{
      color:red;
      font-size:13px;
    }
    .success{
      color:green;
      font-size:14px;
      text-align:center;
    }
    button{
      width:100%;
      padding:10px;
      margin-top:15px;
      background:green;
      color:white;
      border:none;
      cursor:pointer;
    }
  </style>
</head>
<body>

<div class="container">
  <h2>Gym Admission Form</h2>

  <form id="gymForm">

    <label>Full Name</label>
    <input type="text" id="name">
    <span id="nameError" class="error"></span>

    <label>Age</label>
    <input type="number" id="age">
    <span id="ageError" class="error"></span>

    <label>Email</label>
    <input type="email" id="email">
    <span id="emailError" class="error"></span>

    <label>Mobile Number</label>
    <input type="text" id="mobile">
    <span id="mobileError" class="error"></span>
  </form>
</div>

```

```

        <label>Membership Plan</label>
        <select id="plan">
            <option value="">Select Plan</option>
            <option>Monthly</option>
            <option>Quarterly</option>
            <option>Yearly</option>
        </select>
        <span id="planError" class="error"></span>

        <button type="submit">Submit</button>

        <p id="result" class="success"></p>

    </form>
</div>

<script>

// Name Validation (input event)
document.getElementById("name").addEventListener("input", function(){
    let name=this.value;
    if(/^[A-Za-z ]+$/.test(name)){
        document.getElementById("nameError").innerHTML="";
    }else{
        document.getElementById("nameError").innerHTML="Only letters allowed.";
    }
});

// Age Validation (blur event)
document.getElementById("age").addEventListener("blur", function(){
    let age=this.value;
    if(age>=16 && age<=60){
        document.getElementById("ageError").innerHTML="";
    }else{
        document.getElementById("ageError").innerHTML="Age must be between 16 and 60.";
    }
});

// Email Validation (input event)
document.getElementById("email").addEventListener("input", function(){
    let email=this.value;
    let pattern=/^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    if(pattern.test(email)){
        document.getElementById("emailError").innerHTML="";
    }else{
        document.getElementById("emailError").innerHTML="Invalid email.";
    }
});

// Mobile Validation (input event)
document.getElementById("mobile").addEventListener("input", function(){
    let mobile=this.value;

    if(/^\d{10}$/.test(mobile)){
        document.getElementById("mobileError").innerHTML="";
    }else{
        document.getElementById("mobileError").innerHTML="Enter 10-digit mobile number.";
    }
});

// Membership Validation (change event)
document.getElementById("plan").addEventListener("change", function(){
    if(this.value==""){
        document.getElementById("planError").innerHTML="Please select a plan.";
    }else{
        document.getElementById("planError").innerHTML="";
    }
});

```

```
});

// Form Submission (submit event)
document.getElementById("gymForm").addEventListener("submit", function(e){

    e.preventDefault();

    if(
        document.getElementById("nameError").innerHTML==" " &&
        document.getElementById("ageError").innerHTML==" " &&
        document.getElementById("emailError").innerHTML==" " &&
        document.getElementById("mobileError").innerHTML==" " &&
        document.getElementById("plan").value!=" "
    ){
        document.getElementById("result").innerHTML="Gym Admission Successful!";
    }
    else{
        document.getElementById("result").innerHTML="";
        alert("Please correct the errors before submitting.");
    }
});

</script>

</body>
</html>
```

## 4. Output:

### Part A — Gym Admission Registration Form

PRN: 24070521038 | Name: Vansh Mahant | File: /JAVASCRIPT/GYM PRACTICAL 8/index.html

The screenshot shows a web browser window with the address bar displaying the file path: file:///Users/adarshmahant/Desktop/JAVASCRIPT/GYM%20REGISTRATION%20PRACTICAL%208/index.html. The page title is "Gym Admission Registration Form" and the subtitle is "Practical 8: Form Validation & JavaScript Events".

The form contains the following fields and options:

- Name:** Input field with the value "vansh mahant".
- Sex:** Radio buttons for "Male" (selected) and "Female".
- Eye color:** Dropdown menu with the value "brown".
- Check all that apply:** Checkboxes for "Over 6 feet tall" (selected) and "Over 200 pounds".
- Describe your athletic ability:** Text area with the value "I PLAY BADMINTON". A character count "17 characters entered" is shown at the bottom right of the text area.

At the bottom of the form, there are two buttons: "Enter my information" and "Reset".

Below the form, a green box displays a success message: "Registration Submitted Successfully!". Below this message, the submitted information is listed:

- Full Name: vansh mahant
- Sex: Male
- Eye Color: brown
- Attributes: Over 6 feet tall
- Athletic Ability: I PLAY BADMINTON

### Part B — Gym Admission Form

PRN: 24070521038 | Name: Vansh Mahant | File: /JAVASCRIPT/GYM REGISTRATION PRACTICAL/INDEX.html

The screenshot shows a web browser window with a file path in the address bar: file:///Users/adarshmahant/Desktop/JAVASCRIPT/GYM%20PRACTICAL%208/INDEX/. The main content is a 'Gym Admission Form' with the following fields and values:

- Full Name: VANSI MAHANT
- Age: 20
- Email: vanshmahant3@gmail.com
- Mobile Number: 8858896886
- Membership Plan: Quarterly

Below the fields is a green 'Submit' button. At the bottom of the form, a message reads 'Gym Admission Successful!'.

## 5. Result/Conclusion:

Two Gym Admission web forms were successfully designed and implemented using HTML, CSS and JavaScript. Part A uses a table-structured layout with real-time name/textarea validation, a live character counter, and dynamic status messages on the eye-color and attribute-checkbox selections, culminating in a submitted-details summary card. Part B validates name (letters only), age (16–60), email format, 10-digit mobile number, and a required membership plan selection, each tied to appropriate input, blur, and change events, with a success message shown only when every field passes validation. This experiment reinforced form validation techniques and JavaScript event handling (input, blur, focus, change, submit, reset) across two independent implementations.

---

Upload the Soft copy as well as complete code on GitHub link is mandatory.

File name: Practical 8\_24070521038\_Vansh Mahant\_A\_A1